

Rwanda House Price Prediction with Multiple Linear Regression

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1. Introduction and Objective

The project develops a machine-learning system that estimates the selling price of a house in Rwanda from its physical and location characteristics. The target variable is House_Price_Million_RWF. The predictors are Area_m2, Bedrooms, Bathrooms, House_Age_Years, Distance_to_City_km, Parking_Spaces, and Neighborhood.

2. Data Preparation and Cleaning

The supplied cleaning workflow identified 125 original observations. Five exact duplicate records were identified and removed. Nine missing cells were identified. Numerical predictor values were treated with median imputation, Neighborhood with mode imputation, and the target with mean imputation. Potential numerical outliers were assessed using the $1.5 \times \text{IQR}$ rule through sequential filtering. The resulting cleaned dataset contains 91 rows and 9 columns, with zero duplicate rows and zero missing cells.

Problem	Action	Reason
Duplicate records	Remove duplicates	Prevent repeated observations from influencing training
Missing numeric predictors	Median imputation	Retain observations and reduce sensitivity to extreme values
Missing Neighborhood	Mode imputation	Use the most frequent observed category
Missing target	Mean imputation	Retain the observation following the project cleaning workflow
Potential outliers	$1.5 \times \text{IQR}$ filtering	Reduce influence of extreme observations

3. Exploratory Analysis

The exploratory stage examined the distribution of house prices, the relationship between area and price, price variation by neighborhood, and correlations among numeric variables. Bedrooms and Bathrooms were strongly related in the earlier analysis, so multicollinearity was considered when interpreting the regression coefficients.

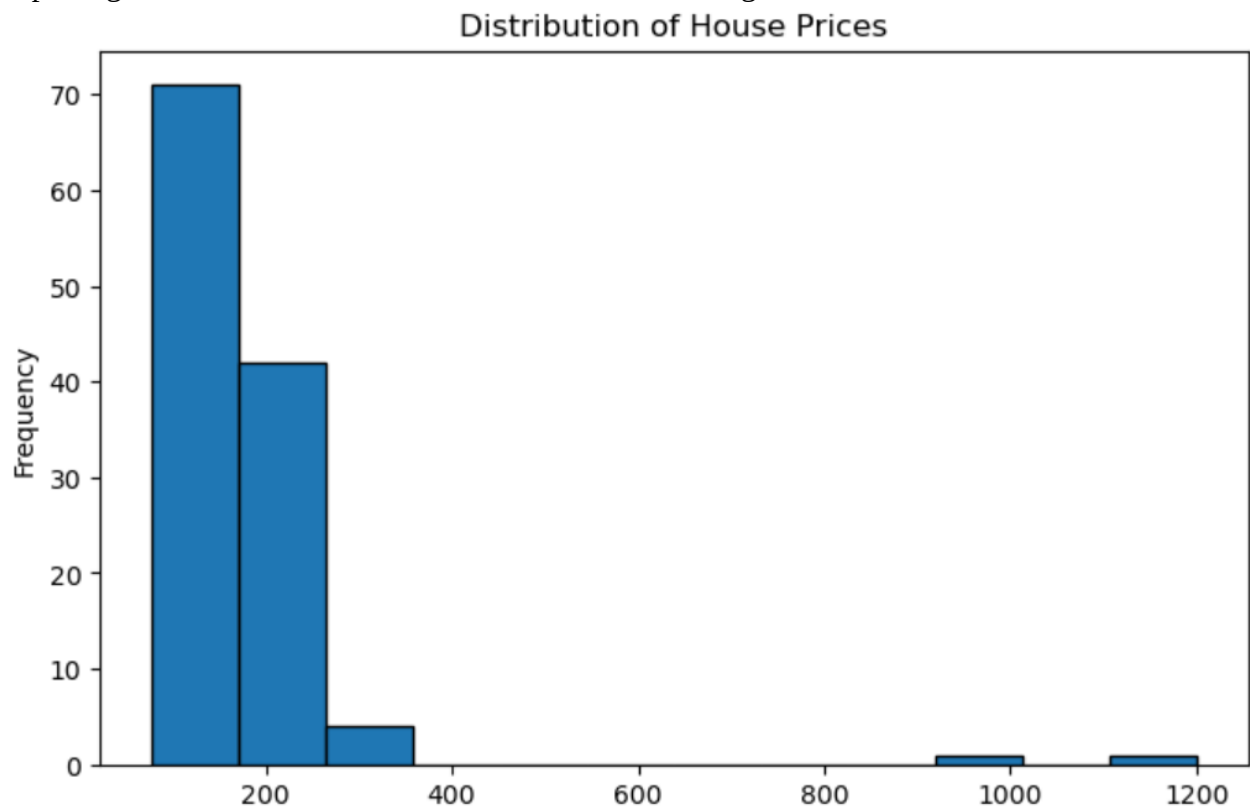


Figure 1. Distribution of house prices.

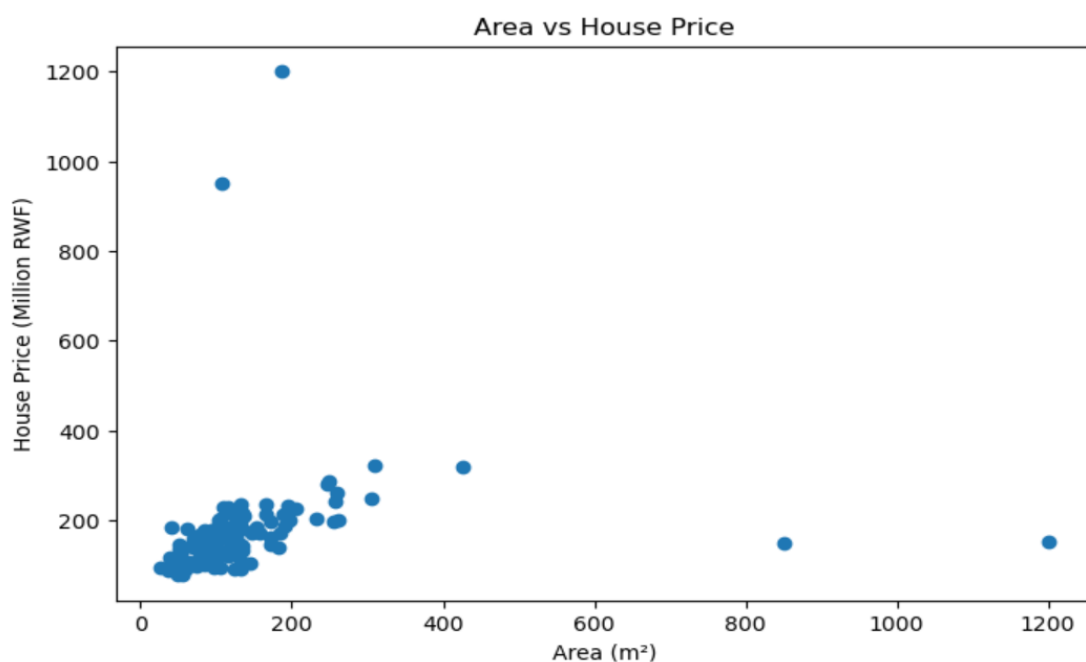


Figure 2. Area versus house price.

4. Model Development

A Multiple Linear Regression model was trained using an 80/20 train-test split with random_state=42. The preprocessing pipeline uses median imputation for numerical variables and one-hot encoding for Neighborhood with one category dropped to avoid the dummy-variable trap. The full pipeline is saved as house_price_model.sav so that raw user inputs can be passed directly to model.predict().

5. Model Performance

Dataset	R ²	Adjusted R ²	MAE (M RWF)	RMSE (M RWF)
Training	0.6554	0.5922	18.16	23.62
Testing	0.6544	0.1113	19.64	22.35

The model explains approximately 65.4% of the variation in the test-set house prices under this split. The test MAE is approximately 19.64 million RWF, and the test RMSE is approximately 22.35 million RWF.

6. Coefficients and Interpretation

Variable	Coefficient
numeric Area_m2	0.4655
numeric_Bedrooms	4.0715
numeric_Bathrooms	14.9252
numeric_House_Age_Years	-0.8615
numeric_Distance_to_City_km	-0.3414
numeric_Parking_Spaces	1.8216
categorical_Neighborhood_Huye	-14.5070
categorical_Neighborhood_Kicukiro	-17.7078
categorical_Neighborhood_Kigali City	3.2575
categorical_Neighborhood_Musanze	-15.8455
categorical_Neighborhood_Nyarugenge	-5.5316

Area_m2 has a coefficient of 0.465, so, holding the other included variables constant, an additional square metre is associated with an estimated increase in price of approximately that amount in million RWF in this fitted model. House_Age_Years has a negative coefficient of -0.862, while Distance_to_City_km has a negative coefficient, indicating negative fitted relationships for those variables in this dataset.

7. Diagnostic Analysis



Figure 3. Actual versus predicted prices.

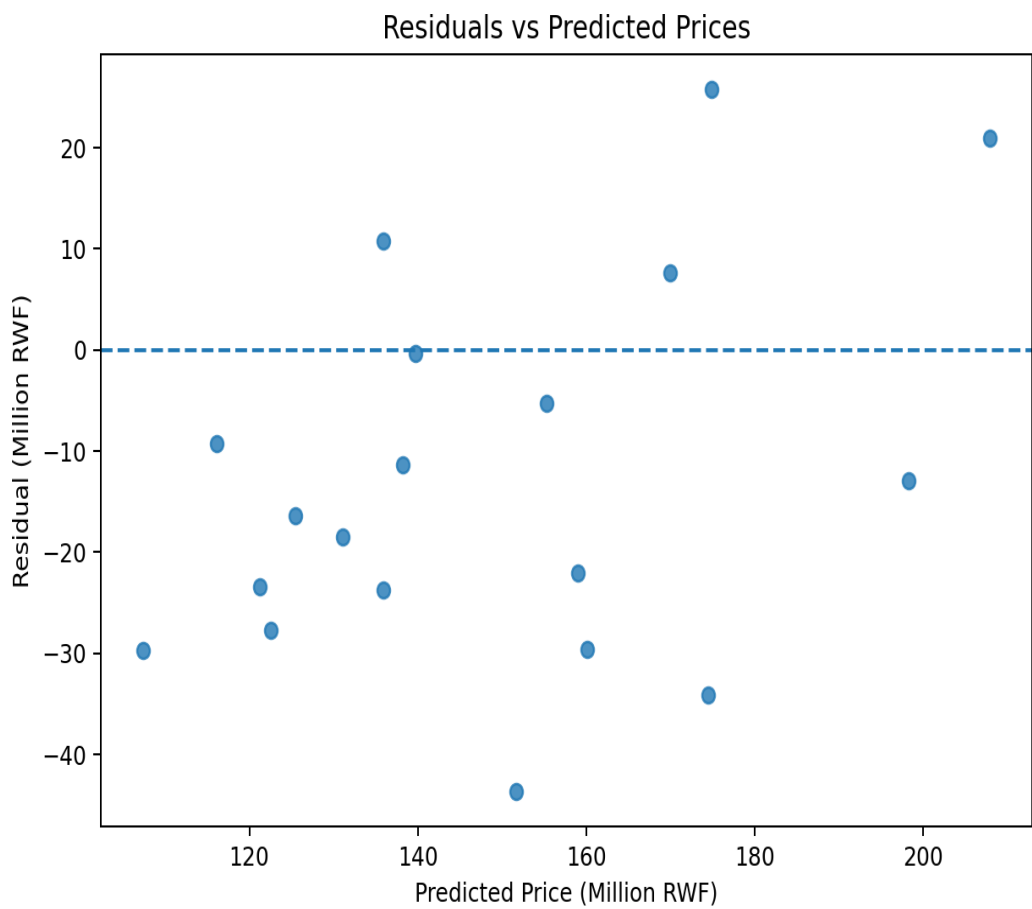


Figure 4. Residuals versus predicted prices.

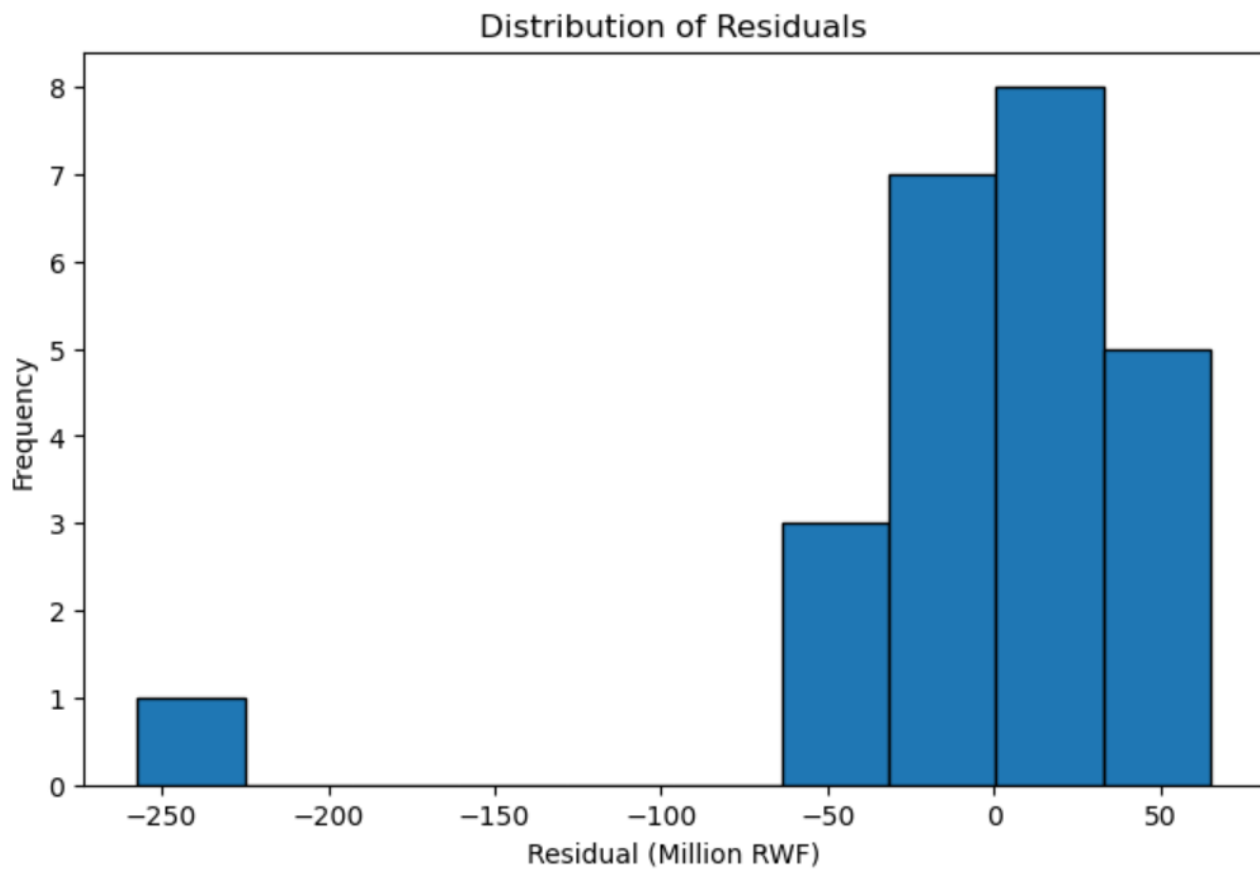


Figure 5. Distribution of residuals.

8. Example Prediction

For a house with area 180 m², 4 bedrooms, 3 bathrooms, age 5 years, 4 km distance to the city centre, 2 parking spaces, and Gasabo neighborhood, the retrained and deployment-compatible model predicts **171.82 million RWF**.

9. Limitations and Conclusion

The dataset is relatively small, so the model may not generalise to all Rwanda housing markets. Multiple linear regression also assumes linear relationships and may not capture nonlinear pricing effects. The model should therefore be treated as an estimation tool rather than a guaranteed market valuation. The project nevertheless provides a complete workflow from cleaned data and regression modelling to a reusable .sav pipeline and a Streamlit prediction application.